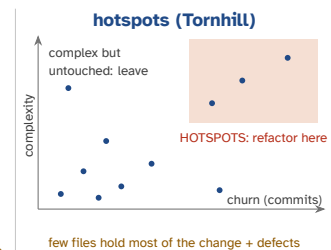
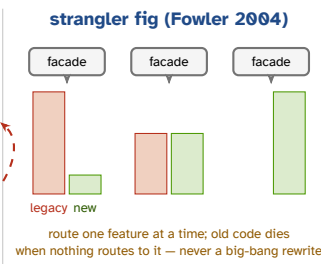
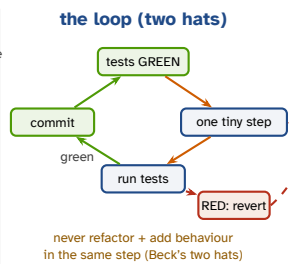
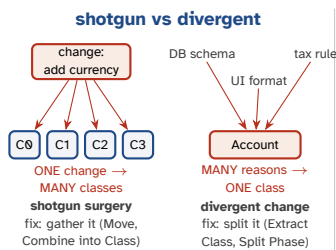


In one line: Refactoring (Fowler): “a change made to the internal structure of software to make it easier to understand and cheaper to modify **without changing its observable behaviour**”. A **smell** is a surface symptom that *suggests* a deeper problem — name the smell, name the refactoring, and keep behaviour pinned: **tests first, tiny steps, commit each green step**.



Smell	Tell-tale	Refactoring (Fowler 2nd ed.) · Swift form
Long Function	scroll to read; comments label the blocks	Extract Function , Decompose Conditional, Split Phase
Large Class / god object	many fields, many reasons to change; Massive VC	Extract Class → ViewModel, data source, Coordinator
Long Parameter List	5+ args; the same group repeats	Introduce Parameter Object , Preserve Whole Object; a struct with defaults
Data Clumps	lat, lon, accuracy always travel together	Extract Class / Parameter Object — the clump is a type
Primitive Obsession	String email, Double money, Int status	Replace Primitive with Object : struct Email validating in init?, Decimal, enum
Feature Envy	a method uses another object's data more than its own	Move Function to the data (tell, don't ask)
Divergent Change	ONE class edited for MANY unrelated reasons	Extract Class , Split Phase (SRP)
Shotgun Surgery	ONE change edits MANY classes	Move Function/Field , Combine Functions into Class, Inline Class
Repeated Switches	the same switch type in many places	Replace Conditional with Polymorphism — or an enum + exhaustive switch (compiler finds every site)
Speculative Generality	protocol with one conformer, unused hooks “for later”	Collapse Hierarchy , Inline Function/Class, Remove Dead Code
Message Chains	a.b.c.d — the caller knows the whole path	Hide Delegate (overdone → Middle Man → Remove Middle Man)
Temporary Field	an optional that is nil except in one flow	Extract Class ; Swift: an enum state with associated values
Refused Bequest	subclass ignores or crashes on what it inherits	Replace Subclass/Superclass with Delegate (compose), Push Down
Comments (deodorant)	a comment explains <i>what</i> the code does	Extract Function with that name, Rename

The discipline No tests → **write characterization tests first** (Feathers: legacy = code without tests): pin *today's* behaviour, bugs included — assert a wrong value, read the actual from the failure, paste it in; snapshot/golden master for big outputs. Then: **one tiny behaviour-preserving step** → run tests → **commit**; red → **revert**, don't debug. **Two hats** (Beck): refactoring *or* adding behaviour, never both in one step/PR. **Preparatory refactoring**: “make the change easy (warning: this may be hard), then make the easy change” (Beck). Prefer IDE refactors (Xcode Rename, Extract to Method) — mechanical = safe. Public interface: **parallel change** (expand → migrate callers → contract), with @available(*, deprecated, renamed:).

Big rewrites — strangler fig Put a facade/router in front; send one feature at a time to the new code; the legacy shrinks until nothing routes to it. iOS: new SwiftUI screens in UIHostingController inside the UIKit app; a protocol in front of the old service with a feature flag choosing the implementation (**branch by abstraction**). Every step ships and can be rolled back. The warning against big-bang: Spolsky, “Things You Should Never Do” (Netscape's rewrite).

```
// BEFORE: primitive obsession + long list + type code
func book(from: String, to: String, date: Date,
adults: Int, kids: Int, cabin: Int, email: String)
// AFTER: value object, parameter object, enum
struct Email { let raw: String
init?(s: String) { guard s.contains("@") else { return nil }
raw = s } }
enum Cabin { case economy, business, first }
struct Trip { var from, to: Airport; var date: Date
var adults = 1, kids = 0; var cabin = Cabin.economy }
func book(_ trip: Trip, contact: Email)
```

Remember Name the smell, name the refactoring · pin behaviour first · one step, green, commit · strangle, don't rewrite · refactor the hotspots.

When NOT to refactor Code nobody needs to change (ugly but stable: leave it — refactor where you *work*; boy-scout rule applies to the area you touch) · no tests and no cheap way to get them · right before a release · a published API without a deprecation path · taste alone. A refactoring needs a *payoff*: the next feature, a bug class, onboarding.

Measure, don't guess Cyclomatic complexity (McCabe, 1976) = $E - N + 2P$ on the control-flow graph $\approx 1 +$ decision points (if, guard, loops, case, &&, ||, ?:, catch); = the minimum number of test paths. McCabe's limit 10; SwiftLint cyclomatic_complexity warns at 10, errors at 20. **Cognitive complexity** (SonarSource) also penalises nesting. **Hotspots** (Tornhill, *Your Code as a Crime Scene*): change frequency from git log × complexity — refactor the top-right, not the whole codebase. Also: SwiftLint type_body_length, function_parameter_count; coverage at the change point.

Traps

- ! **Shotgun surgery** (one change, many classes) ≠ **divergent change** (one class, many reasons) — opposites.
- ! Refactoring without tests is just *changing code*; “refactor” a feature in = two hats.
- ! In Swift a switch over an enum is not the smell — a switch over a *type code* (Int, String) repeated everywhere is.
- ! “Moved the VC's code into a Manager” relocated the smell (god object), not fixed it.